Database design (Entity-Relationship Model)

- how to design and model a DB to store your data
  - what are tables, what are keys, etc

Commonly expressed via diagrams

1976 Peter Chen - Entity-Relnship Model
- wanted to create a unifying framework

→ this is not precisely the <u>relational</u> model
- higher level, a little <u>different</u>

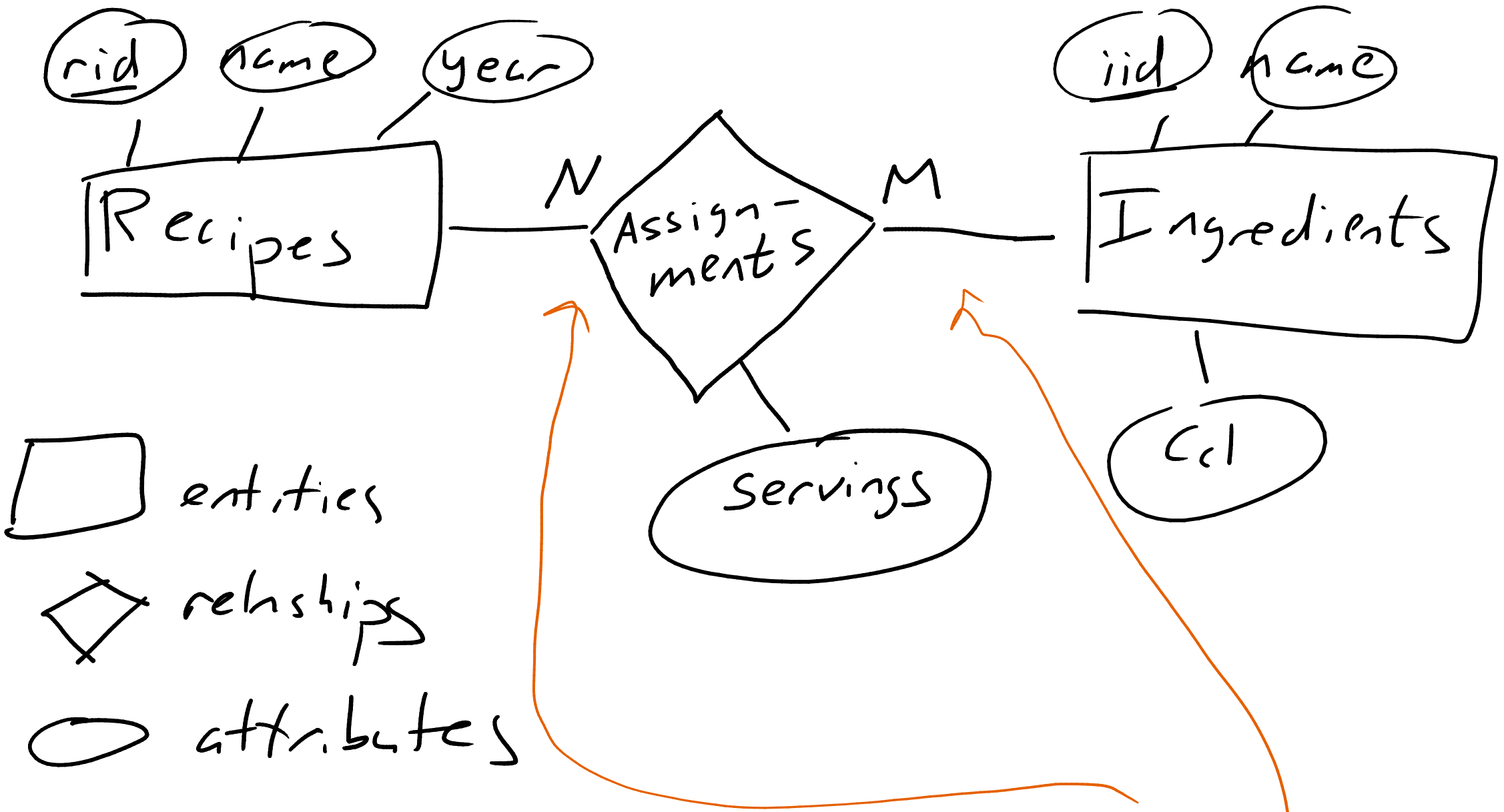
- supposed to be more accessible/understandable by lay people

People use this today, though specifics are allover the place.

One big distinction w/ <u>reln</u> model is that E-R actually has a distinction between entities and relnships

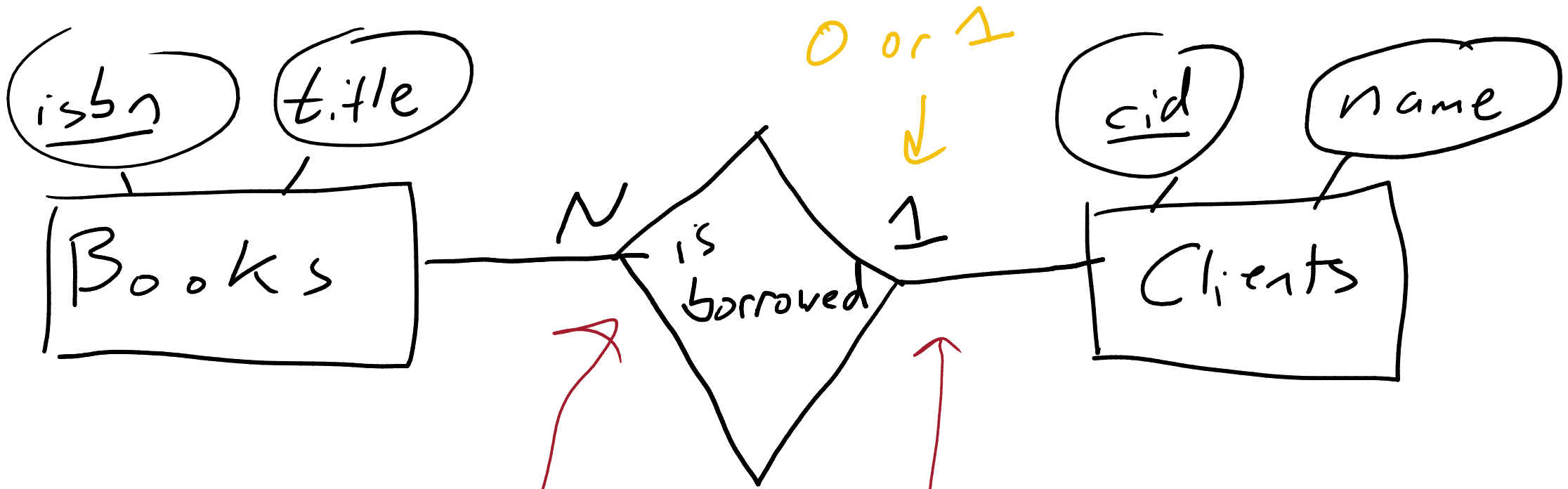
entities → things

relnships → connections



Relnship cardinalities indicate how many times entities paticipate

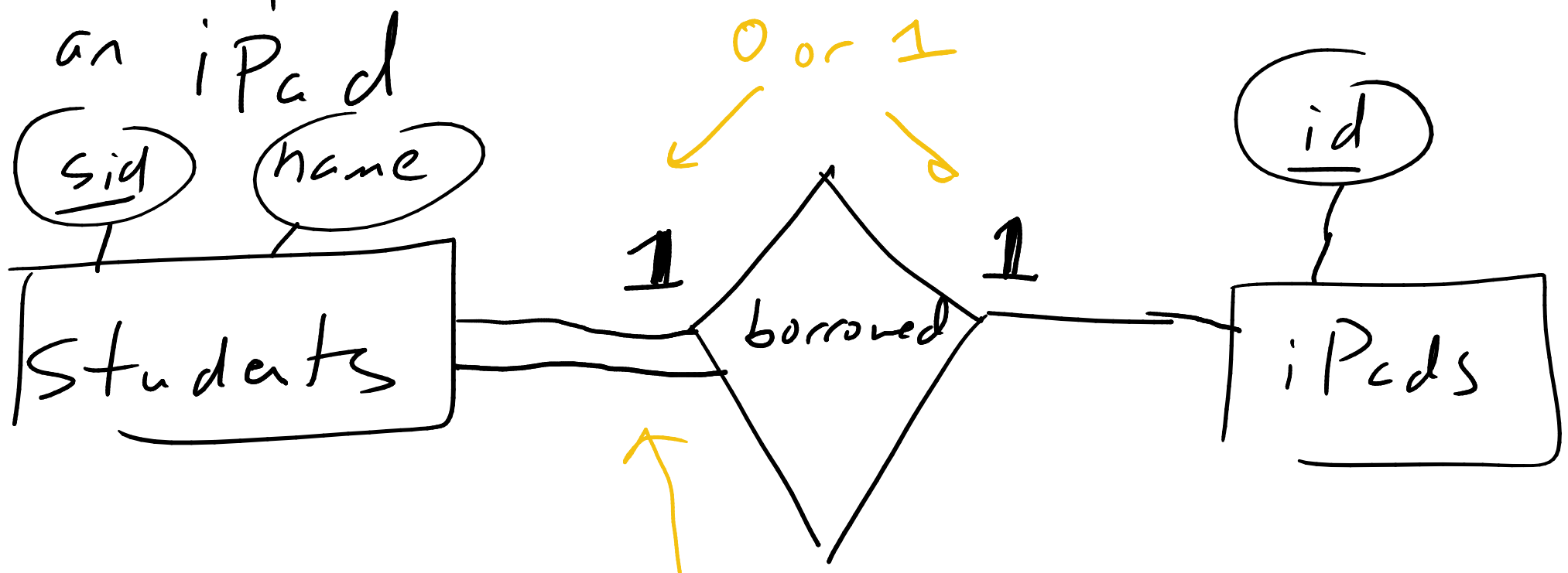
- recipes can have multiple ingreds
- an ingred can be assoc w/ mult recipes



a client can borrow multiple books
a book can only be borrowed by one client

"N books can map to 1 client"
many-to-one relship

N field public schools, every student is issued an iPad



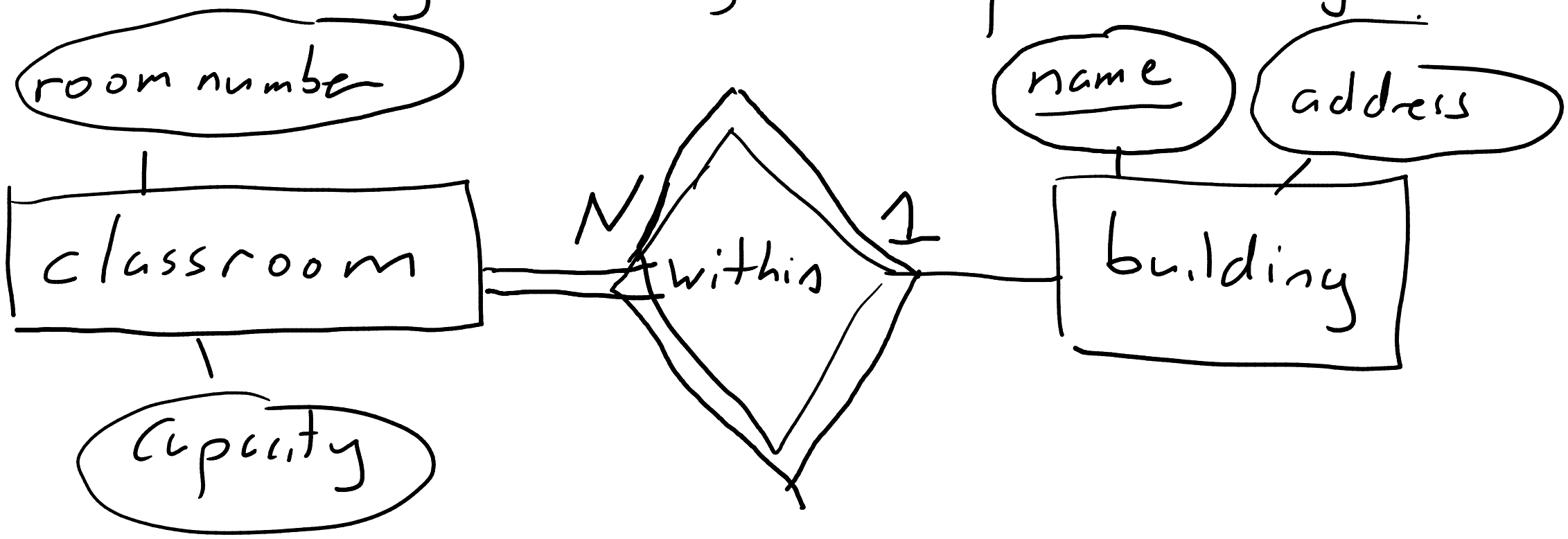
Each students get an iPad

Each iPad maxes at one student

Force a relnship

Every student <u>must</u> participate, meaning every student has an iPad

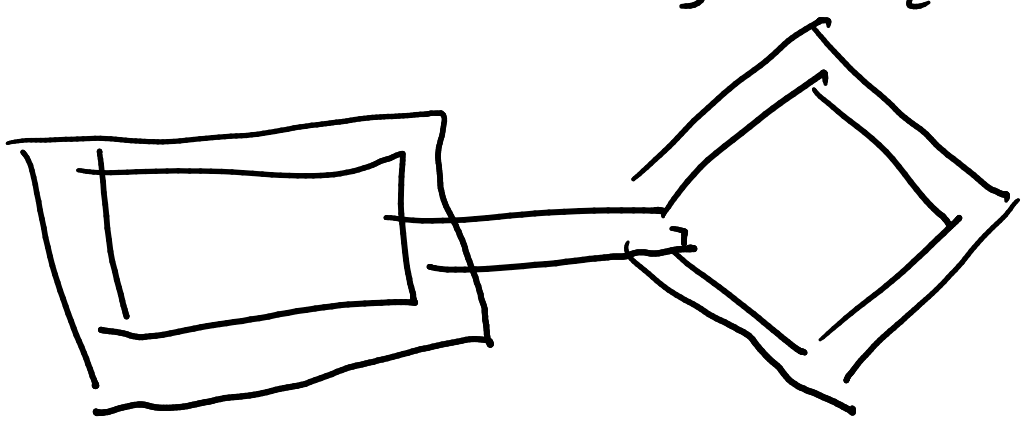
Weak entity - entity w/ a "partial key"



room number does not uniquely identify on its own, but is unique when combined w/ building

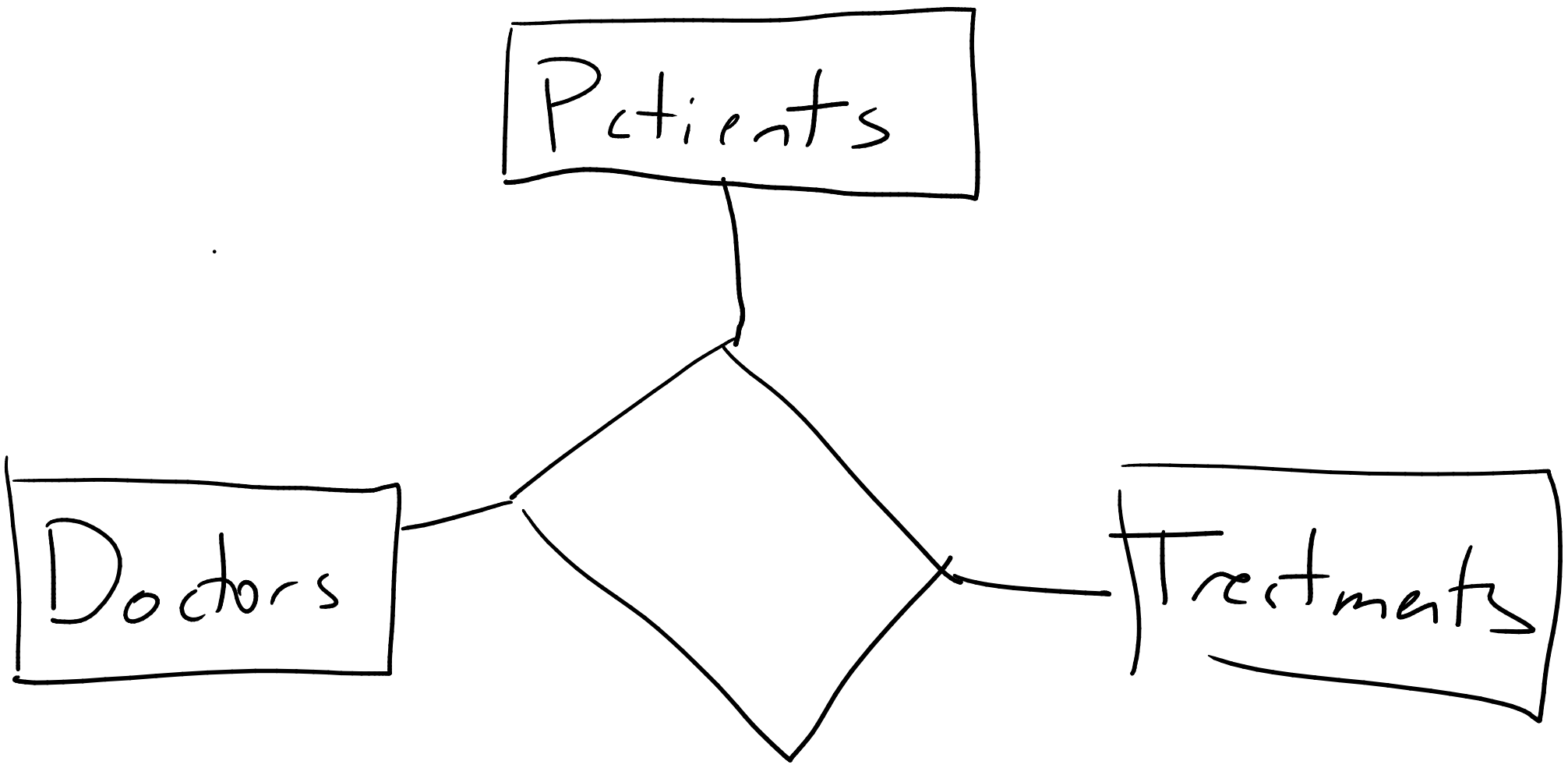
classroom is a weak entity
- not unique on its own
- unique when coupled with a particular relnship

A weak entity requires participation



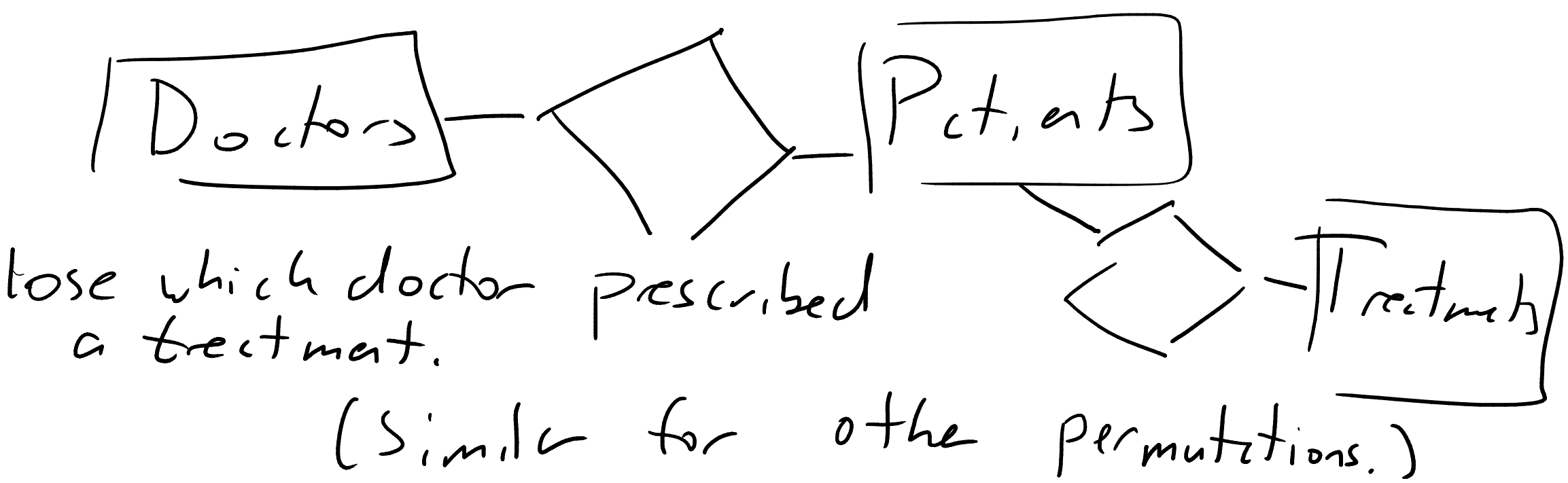
weak entity

Ternary relationships



I am modeling that a doctor prescribes a treatment for a patient

Can you do w/o ternary?



lose which doctor prescribed a treatment.

(Similar for other permutations.)